

P8105_hw1

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Problem1

```
data("penguins", package = "palmerpenguins")
```

The dataset penguins contains variables such as species, islands, bill length, bill depth, flipper length, body mass, sex, and year. The species include Adelie, Gentoo, Chinstrap. The islands include Torgersen, Biscoe, Dream. The dataset has 344 observations and 8 variables. The mean flipper length is 200.9152047mm.

```
# Scatterplot
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.2.1    ✓ readr      2.2.0
## ✓ forcats    1.0.1    ✓ stringr    1.6.0
## ✓ ggplot2     4.0.3    ✓ tibble     3.3.1
## ✓ lubridate  1.9.5    ✓ tidyr      1.3.2
## ✓ purrr      1.2.2
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conf
```

```
penguin_plot = ggplot(penguins, aes(x = bill_length_mm, y = flipper_length_mm, col
```

```
# Export scatterplot
```

```
ggsave("penguin_scatterplot.png", plot = penguin_plot)
```

```
## Saving 7 x 5 in image
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale r
## (\geom_point())
```

Problem2

```
dataframe_P2 = tibble(
  random_sample = rnorm(10),
  logical_vector = random_sample > 0,
  character_vector = c("a", "b", "c", "d", "e", "f", "g", "h", "i", "j"),
  factor_vector = factor(c("1", "2", "1", "1", "2", "3", "3", "2", "1", "3"))
)
```

```
mean(pull(dataframe_P2, random_sample))
```

```
## [1] 0.4678695
```

```
mean(pull(dataframe_P2, logical_vector))
```

```
## [1] 0.5
```

```
mean(pull(dataframe_P2, character_vector))
```

```
## Warning in mean.default(pull(dataframe_P2, character_vector)): argument is not
## numeric or logical: returning NA
```

```
## [1] NA
```

```
mean(pull(dataframe_P2, factor_vector))
```

```
## Warning in mean.default(pull(dataframe_P2, factor_vector)): argument is not
## numeric or logical: returning NA
```

```
## [1] NA
```

The mean can be calculated for the numeric and logical variables, but not for the character and factor variables.

```
as.numeric(pull(dataframe_P2, logical_vector))  
as.numeric(pull(dataframe_P2, character_vector))  
as.numeric(pull(dataframe_P2, factor_vector))
```

The logical variables are converted to 0(FALSE) and 1(TRUE). The character variables are converted to NA because values cannot be interpreted as numbers. The factor variables are converted to integer. This helps explain the previous calculation of mean, logical vector can be treated as 0 and 1, but since character and factor variables are not numeric variables, so they cannot be directly calculated.